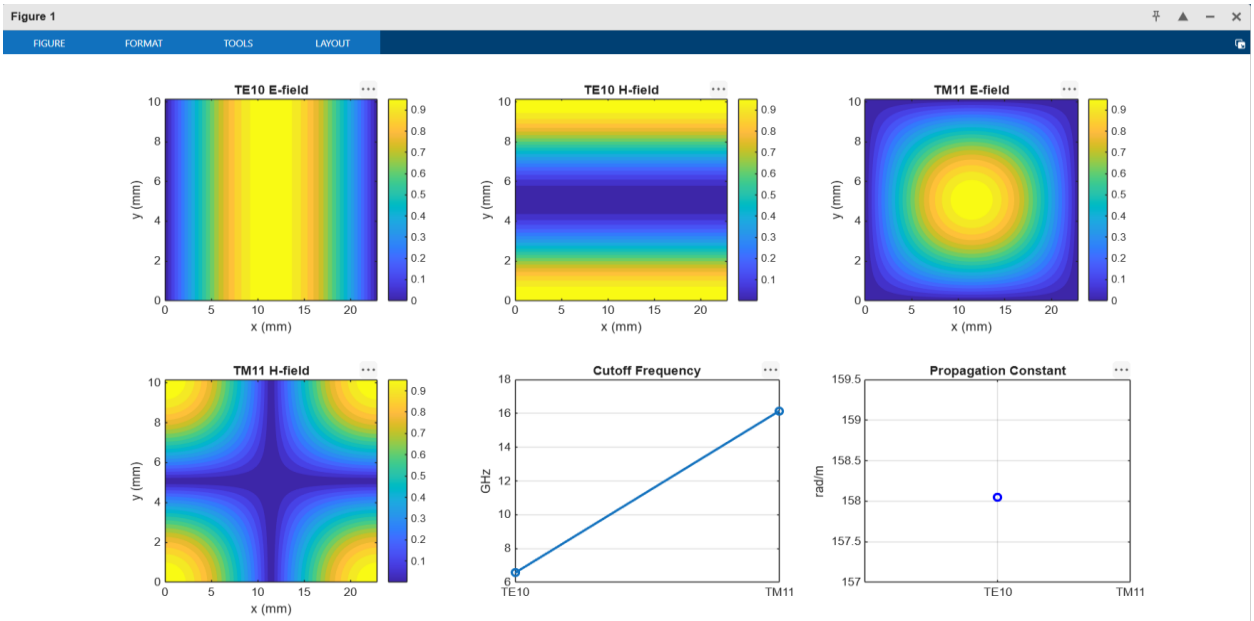
Experiment 1

Code:

```
clc; clear; close all; c=3e8; a=22.86e-3; b=10.16e-3; f=10e9; fc10=c/(2a);
fc11=c/2sqrt((1/a)^2+(1/b)^2); k=2pif/c; beta10=ksqrt(1-(fc10/f)^2); x=linspace(0,a,80);
y=linspace(0,b,60); [X,Y]=meshgrid(x1e3,y1e3); E10=sin(piX/(a1e3));
H10=0.5(1+cos(2piY/(b1e3))); E11=sin(piX/(a1e3)).sin(piY/(b1e3));
H11=abs(cos(piX/(a1e3)).cos(piY/(b*1e3))); figure('Color','w','Position',[100 50 1400 700]);
subplot(2,3,1); contourf(X,Y,E10,20,'LineColor','none'); title('TE10 E-field'); xlabel('x (mm)');
ylabel('y (mm)'); colorbar; subplot(2,3,2); contourf(X,Y,H10,20,'LineColor','none'); title('TE10
H-field'); xlabel('x (mm)'); ylabel('y (mm)'); colorbar; subplot(2,3,3);
contourf(X,Y,E11,20,'LineColor','none'); title('TM11 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
colorbar; subplot(2,3,4); contourf(X,Y,H11,20,'LineColor','none'); title('TM11 H-field'); xlabel('x
(mm)'); ylabel('y (mm)'); colorbar; subplot(2,3,5); plot([1 2],[fc10 fc11]/1e9,'-o','LineWidth',2);
set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); title('Cutoff Frequency'); ylabel('GHz');
grid on; subplot(2,3,6); plot(1,beta10,'bo','LineWidth',2); set(gca,'XTick',[1
2],'XTickLabel',{'TE10','TM11'}); title('Propagation Constant'); ylabel('rad/m'); ylim([157
159.5]); grid on; fprintf('TE10 cutoff = %.3f GHz\n',fc10/1e9); fprintf('TM11 cutoff = %.3f
GHz\n',fc11/1e9); fprintf('TE10 beta = %.3f rad/m\n',beta10); fprintf('TE10 propagates; TM11 is
below cutoff at 10 GHz.\n');
```

Output:



TE10 cutoff = 6.562 GHz

TM11 cutoff = 16.156 GHz

TE10 beta = 158.046 rad/m

TE10 propagates; TM11 is below cutoff at 10 GHz.

